

DELHI PUBLIC SCHOOL, BOKARO STEEL CITY

Class X - Science (Biology) - Unit Test

Time: 1 Hour

Maximum Marks: 50

Answer all questions. Draw diagrams wherever required.

1. Which blood vessel carries blood away from the heart? [2]
2. Which of the following organelles is primarily involved in photosynthesis? [2]
3. Explain the role of chloroplasts in photosynthesis, naming the main pigments involved and briefly outlining the two major stages of the process. [2]
4. Describe the flow of blood through the human heart starting from the right atrium and ending at the aorta; include the names of valves crossed. [2]
5. Draw a labelled diagram of an alveolus showing capillaries and air space (label alveolar sac, capillary, and direction of gas exchange). [2]
6. Draw a neat labelled diagram of the human digestive system (stomach, small intestine, large intestine, liver, pancreas) and label the site where most absorption occurs. [5]
7. Draw and label a nephron (Bowman's capsule, glomerulus, proximal tubule, loop of Henle, distal tubule, collecting duct). [5]
8. Explain the structural differences between palisade mesophyll and spongy mesophyll and state how each structure aids its function in the leaf. [5]
9. Describe the process of transpiration in plants in two to three sentences and name two environmental factors that increase its rate. [5]
10. Explain how the structure of xylem vessels facilitates water transport in plants (mention one structural feature and its role). [5]
- 11 (a). A diagram shows two potted plants - Plant A in bright light with broad green leaves, Plant B kept in dim light with pale, elongated leaves. State which plant is healthier and give one reason. [2]
- 11 (b). Suggest one practical measure to help Plant B recover. [3]
- 12 (a). A resting person has a tidal volume (air per breath) of 0.5 L and breathes 12 times per minute. Calculate the pulmonary ventilation per minute. [5]
- 12 (b). If dead space is 0.15 L per breath, calculate the alveolar ventilation per minute. Show working. [5]